Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 4 Marks

SOLUTION REQUIREMENTS SPECIFICATION

Functional Requirements (FR)

The functional requirements describe the system capabilities that the underwriting software must provide.

FR ID	Feature Name	Detailed System Requirement Specification
FR-1	Underwriting Input Form	The system must render a web page capturing 12 parameters: Gender, Car, Realty, Income Type, Education, Family Status, Housing Type, Occupation, Annual Income, Age, Employment Duration, and Family Members.
FR-2	Policy Override Engine	Before calling the ML model, the server must evaluate four policy filters: Minimum Income Floor ($15k), Unemployed Student (0 yrs employed), Underage Unemployed (Age<=18 and 0 yrs employed), and Low-Income Student (<$20k). If any trigger, immediately reject application.
FR-3	Feature Preprocessing	If policies pass, the system must compute engineered ratios (AGE_YEARS, EMPLOYMENT_YEARS, INCOME_PER_MEMBER, EMPLOYMENT_RATIO, IS_EMPLOYED) and scale them using the pre-fitted StandardScaler binary.
FR-4	ML Model Inference	The system must load the pre-trained Random Forest model (credit_model.pkl) and check if default probability >= threshold.pkl (0.2312) to classify as REJECTED, else APPROVED.
FR-5	Outcomes Panel	The dashboard must display a green APPROVED or red REJECTED banner, the exact risk probability, detailed logs showing the decision path, and a summary of inputs.

Non-Functional Requirements (NFR)

The non-functional requirements define the operational bounds, performance, and security constraints.

NFR ID	Quality Attribute	Specification & Operational Threshold
NFR-1	Performance & Latency	The end-to-end pipeline (form submission, feature scaling, model scoring, and page rendering) must complete in under 100 milliseconds under local server conditions.
NFR-2	Robustness & Security	The form must validate numerical bounds. The server must omit Personally Identifiable Information (PII) such as Names or SSNs in logs, maintaining compliance with consumer privacy frameworks.
NFR-3	Compliance & Portability	The application must bypass restricted C-compiled libraries (e.g. XGBoost DLLs) that trigger local endpoint computer policy blocks, using scikit-learn's Random Forest which runs natively.
NFR-4	Reproducibility	The model training and scaling pipeline must yield identical classifier outputs across executions by pinning the random seed (random_state=42).

User Stories & Acceptance Criteria

User Story ID	User Role	As a... I want to... So that...	Acceptance Criteria
US-01	Credit Underwriter	As an underwriter, I want the system to filter out obviously high-risk applications instantly, so that I don't waste time on manual review.	Applications with income < $15k or unemployed students must be flagged and rejected by Layer 1 policies in <10ms.
US-02	Risk Manager	As a risk manager, I want the model to handle dataset class imbalance and optimize default captures, so that we minimize credit losses.	Model training must incorporate SMOTE oversampling and threshold calibration, maximizing F1-score on validation split.
US-03	Compliance Auditor	As an auditor, I want to review the exact decision path for every applicant, so that I can ensure compliance with banking rules.	The results screen must display detailed execution logs stating whether a policy rule triggered or if ML inference completed.